Conductivity

Definition : the property of a material that allows it to easily transmit either heat (thermal conductivity) or electricity (electrical conductivity)

Electrical conductivity :

Electrical conductivity measures the extent to which a material allows the flow of electric charge through it when a potential difference (voltage) is applied across it.



Materials are classified based on their electrical conductivity:

- Conductors: Materials with high conductivity that allow electric current to pass through easily (e.g., silver, copper, gold, aluminum). These materials have many free charge carriers.
- Insulators: Materials with very low conductivity that resist the flow of electric current (e.g., plastic, rubber, glass, dry wood). Their electrons are tightly bound to atoms and cannot move freely.
- Semiconductors: Materials with conductivity between that of conductors and insulators (e.g., silicon, germanium). Their conductivity can be significantly altered by temperature or adding impurities (doping).

Conductivity

Thermal Conductivity :

It's the rate of heat flow per unit area per unit temperature gradient (the slope of temperature with distance) for a specific material. This property is extremely important when creating something such as cooking utensils and other products that will be hot and handle by people. This must be taken into account to ensure people don't get burned by the heat from the pan going into the handle.

High Thermal conductivity materials








Low Thermal conductivity materials












Key factors that can influence the thermal conductivity of a material

1. Temperature: Affects metals (conductivity generally decreases with T) and nonmetals (behavior varies) differently, impacting electron-phonon scattering.
2. Density & Porosity: Higher density often means better conductivity (more atoms/electrons); porosity (air pockets) significantly lowers it, making insulators porous.
3. Chemical Composition & Impurities: Alloying or adding impurities changes the material, affecting electron/phonon movement.
4. Structure & Microstructure:
    1. Crystal Structure: Cubic crystals are isotropic, while others (e.g., non-cubic) are anisotropic (different conductivity in different directions).
    2. Grain Size: Larger grains often increase thermal conductivity.
5. Phase & Moisture Content: Phase changes (ice to water) drastically alter conductivity; moisture in porous materials increases it by filling air gaps.
6. Electrical Conductivity: In metals, higher electrical conductivity generally means higher thermal conductivity (Wiedemann-Franz law) due to shared free electrons.
7. Airflow & Pressure: Air velocity and external pressure can impact measurements, especially in porous materials or gases.